ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus
(From Academic Session 2024-25 onwards)

B. TECH
POWER ELECTRONICS AND INSTRUMENTATION ENGINEERING

4th SEMESTER

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure

(From Academic Session 2024-25 onwards)

B. Tech 4th Semester: Power Electronics and Instrumentation Engineering
Semester IV

Sl. No.	Course Code	Course Type	Course Title	Hours per Week			Credit	Marks	
				L	T	P	C	CE	ESE
Theory									
1	PE241401	PCC	Transducers and Sensors	3	0	2	3	30	70
2	PE241402	PCC	Analog Circuits	3	0	2	3	30	70
3	PE241403	PCC	Electrical Machines	3	0	2	3	30	70
4	PE241404	PCC	Python Programming	3	0	0	3	30	70
5	PE241405	PCC	Signals and Systems	3	0	0	3	30	70
6	HS241406	HSMC	Finance and Accounting	3	0	0	3	30	70
7	AU241407	AU	Environmental Science	3	0	0	0 (PP/NP)	100	-
Practical									
8	PE241411	PCC	Transducers and Sensors Laboratory	0	0	2	1	15	35
9	PE241412	PCC	Analog Circuits Laboratory	0	0	2	1	15	35
10	PE241413	PCC	Electrical Machines Laboratory	0	0	2	1	15	35
11	PR241415	PR	Micro Project (Skill Based)	0	0	4	2	15	35
TOTAL				**21**	**0**	**16**	**23**	**340**	**560**
Total Contact Hours per week: 37									
Total Credit: 23									

Course Code	Course Title	Hours per week L-T-P	Credit
PE241401	**Transducers and Sensors**	**3-0-2**	**3**

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1**: Identify different types of transducers
- **CO2**: Understand the characteristics of different transducers.
- **CO3**: Understand the basic principle of various (types) resistive/capacitive/piezoelectric /elastic transducers
- **CO4**: Select appropriate transducers for a specific application.
- **CO5**: Apply transducers for measurement of parameters like temperature, force, pressure, level etc.

MODULE 1: **Introduction** **(3 lectures)**

Classification of Transducers, Characteristics Static characteristics - Accuracy, precision, Dynamic characteristics - Mathematical model of transducer - Zero, 1st and 2nd order system response to impulse, step, ramp and sinusoidal inputs-Error analysis, statistical methods-Odds and uncertainty improvement of signal-to-noise ratio- Calibration methods, Static calibration Selection of transducers.

MODULE 2: Passive transducers **(8 lectures)**

Resistive transducers-Potentiometers RTD-Thermistors- Strain Gauge-Hot wire anemometers and their applications in measurement of pressure, temperature, force; torque, vibration and flow etc.

Capacitive transducer -Capacitive transducer and types - Air gap and dielectric types and their Applications, Capacitor microphone - Frequency response.

Inductive transducers -Variable reluctance transducers - Principle of operation, construction details, LVDT- Principle of operation, construction details, applications. Synchros- Principle of operation, characteristics and application.

MODULE 3: Active transducers **(6 lectures)**

Piezoelectric transducers: Piezoelectric crystal and its properties, working principle- Sensitivity coefficients, Applications.

Thermocouples: Laws of thermocouple, Fabrication of industrial thermocouples, Signal conditioning of thermocouple output-Isothermal block reference junctions, Commercial circuits for cold junction Compensation, Response of thermocouple, special techniques for measuring high temperature using Thermocouples.

MODULE 4: Optical transducers **(5 Lectures)**

Photo detectors-LDR. Photo diode, Photo transistor- Photo-emissive cell - Photoconductive cell - Photovoltaic cells- Radiation and optical pyrometers.

MODULE 5: Elastic transducers **(3 Lectures)**

Diaphragms--Capsules- Bourdon Tube-Bellows-Dynamic Considerations-Applications,

MODULE 6: Electro-mechanical Transducers **(9 lectures)**

D.C and A.C Servo Motors -Construction, working principle, torque equation, transfer functions and applications of d.c and a,c servo motors.

Stepper motors-Classification- Construction - Principle of operation -Torque equation Driver circuits - Applications.

MODULE 7: Special Transducers **(6 lectures)**

D.C & A.C Techo generators -Construction, Principle of operation, characteristics and applications. Digital transducers- Shaft encoders; Ultrasonic transducers- Principle of operation and applications. Hall Effect transducer, Gyroscope, Smart Sensors- various industrial types, applications.

Suggested Text Books:

1. Process Control-Principles and Applications - Surekha Bhanot, Oxford press.
2. Control Systems-Theory and Applications -Smarajit Ghosh.
3. Transducers and Instrumentation - Murthy D.V.S., PHI. New Delhi.
4. Instrumentation, Measurement and Feed Back - Jones B.E., TMH. New Delhi.
5. Instrumentation Devices and Systems - Ranga, Sarma, Mani; TMH.
6. Instrument Engineer's Hand Book (Measurement) - Liptak B.G, CRC Press
7. Fractional H.P Electrical Machines - Armensky E.V & Falk G.B - (Mir Publishers) 10. Instrumentation measurement and Analysis - Nakra, B.C., and Chaudry, K.K., TMH. 11. Instrument transducers: An introduction to their performance and design – Neubert M.K.P. Clarendon Press.

Reference Books:

1. Electrical and electronics measurements and measuring instruments-A.K Shawney, Dhanpat Rai
2. Measurement Systems: 'Application and Design - Doebiin, McGraw Hill.
3. Transducers and Instrumentation -- Murthy D.V.S... PHI. New Delhi.
4. Sensors and Transducers- Patranabis D., Wheeler.
5. Instrumentation Engineer's Hand Book (Measurement) - Liptalk.

Course Code	Course Title	Hours per week L-T-P	Credit
PE241402	**Analog Circuits**	**3-0-2**	**3**

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1**: Analyze and design single-stage and multistage amplifiers, focusing on small signal analysis, voltage gain, and frequency response, using transistor models and current mirrors.
- **CO2**: Evaluate feedback topologies and their effects on gain, bandwidth, and stability, including gain and phase margin calculations.
- **CO3**: Design and analyze RC and LC oscillators using the Barkhausen criterion for stable oscillations.
- **CO4**: Analyze and design differential amplifiers, calculating gain, CMRR, and applying OP-AMP design principles and compensation techniques.
- **CO5**: Design OP-AMP circuits for linear and non-linear applications, including filters, precision rectifiers, and signal generators.

MODULE 1: Amplifiers (14 Lectures)

Amplifier Models:Voltage amplifier, current amplifier, trans-conductance amplifier and trans-resistance amplifier.Single Stage Amplifier:Small signal analysis, low frequency transistor models, estimation of voltage gain, input resistance, output resistance, High frequency transistor models, high frequency response of single stage amplifier. Current mirror: Basic topology and its variants, V-I characteristics, output resistance and minimum sustainable voltage (VON), maximum usable load.Multistage Amplifiers:Classification, Distortion, Low and high Frequency responses and Bode plots, Cascode amplifiers (CE-CB), effect of coupling and bypass capacitors, RC coupled amplifier and its low frequency response.Power Amplifiers:Various classes of operation (Class A, B, AB and C), their power efficiency and linearity issues.

MODULE 2: Feedback Topologies (4 Lectures)

Voltage series, current series, voltage shunt, current shunt, effect of feedback on gain, bandwidth etc., calculation with practical circuits, concept of stability, gain margin and phase margin

MODULE 3: Oscillators (4 Lectures)

Barkhausen criterion, RC oscillators (phase shift, Wienbridge), LC oscillators (Hartley, Colpitt, Clapp), crystal oscillator.

MODULE 4: Differential Amplifier (4 Lectures)

Basic structure and principle of operation, calculation of differential gain, common mode gain, CMRR and ICMR. OP-AMP design: Ideal OPAMP characteristics and limitations, Design of differential amplifier for a given specification, design of gain stages and output stages, compensation.

MODULE 5: OP-AMP Applications (12 Lectures)

Linear and Non-linear applications: Review of inverting and non-inverting amplifiers, integrator and differentiator, summing amplifier, difference amplifier, instrumentation amplifier, Voltage to current converter, currentto Voltage converter, Comparator, Window detectors,Sample and Hold Circuits, Peak Detectors, Precision rectifier, Log and antilog amplifiers, Schmitttrigger anditsapplications. Active Filters:Low pass, high pass, band pass and band stop.Signal Generators: Sine wave, Square wave, Triangular Wave, Saw tooth wave

Suggested Text Books:

1. A.S. Sedra and K.C. Smith, “Microelectronic Circuits”, Saunder's College11 Publishing, Edition IV
2. J.V. Wait, L.P. Huelsman and GA Korn, “Introduction to Operational Amplifier theory and applications”, McGraw Hill, 1992.
3. J. Millman and A. Grabel, “Microelectronics”, 2nd edition, McGraw Hill, 1988.
4. P. Horowitz and W. Hill, “The Art of Electronics”, 2nd edition, Cambridge University Press, 1989.

Suggested Reference Books:

1. Paul R. Gray and Robert G.Meyer, “Analysis and Design of Analog Integrated Circuits”, John Wiley, 3rd Edition
2. Jacob Millman, Christos Halkias & Chetan Parikh, “Millman’sIntegrated Electronics”, Mc Graw Hill
3. Baker Lee Boyce book for this course need to be recommend as reference/text book

Course Code	Course Title	Hours per week L-T-P	Credit
PE241403	**Electrical Machines**	**3-0-2**	**3**

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1**: Relate the principles of electromagnetic energy conversion and electromagnetic induction to operation of electrical machines.
- **CO2**: Solve simple numerical problems of D.C. machines (both motors and generators).
- **CO3**: Determine losses of a transformer and its efficiency at various load conditions
- **CO4**: Explain the working of an induction motor and solve simple numerical problems.
- **CO5**: Explain the working principle and applications of servo motor, stepper motor and BLDC motor.

MODULE 1: Electromagnetic Induction and Electromechanical Energy Conversion **(2 Lectures)**

Review of magnetic circuits, MMF, Faraday's Law of Electromagnetic induction, magnetically coupled circuits Basic principles of energy conversion, dynamically induced e.m.f. and torque in rotating machines.

MODULE 2: D.C. Machines

(i) **Generators:** Constructional features, Types of Armature windings, Methods of excitations shunt, series and compound. E M F equation, Armature reaction, Characteristic of generators, Losses, Efficiency and Regulation, Parallel operation. **(6 Lectures)**

(ii) **Motors:** Torque equation of motors, Speed and Torque characteristic curves of shunt, series and compound motors, Losses and efficiency. Starting of D C motors and Starters. Speed control-conventional methods and solid-state control, Choice of motors for different duties. **(5 Lectures)**

MODULE 3: Transformers **(10 Lectures)**

Principles of operation of transformer, voltage and current ratios, Construction – shell type and core type, single phase and poly phase, cooling methods, E.m.f. equation. Transformer circuit parameters, equivalent circuit, Phasor diagrams on no-load and on load. Open circuit and short tests. Regulation, losses and efficiency, maximum efficiency, all-day efficiency. 3-phase transformer, Auto-transformer, and instrument transformer.

MODULE 4: A.C. Motors **(12 Lectures)**

3 phase Induction motor: Constructional features (frame, magnetic circuit, stator and rotor electric circuits) and working principle - production of rotating magnetic field and development of torque. Slip speed and slip, torque equations. Torque-slip characteristic. Effect of variation of applied voltage and rotor resistance. Power stages, losses and efficiency. Starting and speed control. Single phase induction motors: principle, double revolving field theory. Starting arrangement of single phase motor and its applications. Comparison between single phase and three phase motors. Brief Introduction of Synchronous Motor.

MODULE 5: Special Motors **(8 Lectures)**

Introduction to special Machines- Servo Motor, Linear Induction Motors, Stepper Motors, Brushless DC Motors and their applications in Instrumentation Systems.

Suggested Text Books:

1. Nagrath D.P. & Kothari I.J, " Electrical Machines", Tata McGraw Hill Education
2. Bimbra, P.S., "Electric Machinery", Khanna Publishers
3. Mehta V.K. and Mehta, R. "Principle of Electrical Machines", S. Chand and Co.

Reference Books:

1. Langsdrof A.S: Theory of Alternating Current Machinery, McGraw Hill Education

2. Chapman.J, “Electric Machinery Fundamentals”, McGraw Hill Book Co.
3. Fitgerald, A.E., Charles Kingsely Jr. Stephen D.Umans, “Electric Machinery” McGraw Hill Books Company
4. J.B. Gupta, "Theory & Performance Of Electrical Machines" Katsons Books

Course Code	Course Title	Hours per week L-T-P	Credit
PE241404	**Python Programming**	**3-0-0**	**3**

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1**: Recall and describe basic concepts of Python programming such as data types, operators, control statements, and strings.
- **CO2**: Explain the use of functions, modules, file handling, and exception handling concepts in Python programming.
- **CO3**: Utilize Python standard libraries and built-in functions for engineering problem-solving.
- **CO4**: Apply Python programming concepts for file handling, exception handling, and data manipulation using collection types.
- **CO5**: Develop object-oriented programs using classes, inheritance, and polymorphism for real-time applications.

MODULE 1: Python Basics **(8 Hours)**

Important features, Identifiers, Variables, operators, Data types. Control Statements: If, If - else, Nested If - else, Loop: for, while, nested loops, break, continue, pass String Operations: Accessing strings, basic operations, string functions.

MODULE 2: Functions and Modules **(8 Hours)**

Defining a function, Calling a function, Types of functions, Function Arguments, Anonymous functions, Global and local variables.

Modules: Importing module, Math module, Random module, Packages, Composition Input-Output: Opening and closing file, Reading and writing files Exception Handling: Exception Definition, Exception Handling, Except clause, Try, finally clause, User Defined Exceptions

MODULE 3: Libraries and collections **(7 Hours)**

Libraries- Introduction, standard libraries- numpy, matplotlib, pandas,math, random etc; built in functions-input(), print(), append(), read(), min/max(), len() etc. Regular Expressions-working, applications;Collection data types-List,Tuple, String,Set,Dictionary.

MODULE 4: File and exception handling **(7 Hours)**

Types of files-csv,txt,img, pdf,binary etc, file operation-Opening/Closing a file,Read/Write file,conversion;Exception -Built-in Exception in Python,Built-in Exception in Python.

MODULE 5: Object Oriented Programming **(6 Hours)**

Class and objects, Inheritence, Polymorphism -Built-in Polymorphism, Functions and Classes, Applications.

Suggested Text Books:

1. Zero To Mastery In Python Programming, by Monu Singh Rakesh K. Yadav, Srinivas Arukonda
2. Python Programming : Beginners Guide by Sundarrajan M, Mani Deepak Choudhry, Jeevanandham S, Akshya Jothi.
3. Ultimate Python Programming: Learn Python with 650+ Programs, 900+ Practice Questions, and 5 Projects by Deepali Srivastava

Reference Books:

1. Learning with Python Paperback by Allen Downey, Jeffrey Elkner, Chris Meyers.
2. Beginning Programming with Python for Dummies, by John Paul Mueller.

Course Code	Course Title	Hours per week L-T-P	Credit
PE241405	**Signals and Systems**	**3-0-0**	**3**

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1**: Identify various types of signals in continuous-time and discrete-time domain
- **CO2**: Understand Linear Time Invariant (LTI) system and its properties to obtain the response of the system using convolution sum and convolution integral
- **CO3**: Apply knowledge of Fourier Series and Fourier Transform to obtain the frequency domain representation and analysis of signals and systems
- **CO4**: Apply sampling techniques for processing of signals
- **CO5**: Apply knowledge of Z transform and LTI system to design and realize digital filters: FIR and IIR filters

MODULE 1: Introduction to Signals and Systems (6 Lectures)

Definitions, continuous-time (CT) and discrete-time (DT) signals, exponential and sinusoidal signals, signal energy and power, eve and odd signals, periodic signals, transformation of independent variables: time-shift, time-reversal and time-scaling, CT and DT systems and their classification, basic properties of CT and DT systems.

MODULE 2: LTI Systems (6 Lectures)

DT LTI systems: convolution sum, CT LTI systems: convolution integral, properties of LTI systems: commutative, distributive and associative properties, LTI systems with and without memory, invertibility, causality and stability of LTI systems, systems described by differential and difference equations, block diagram of LTI systems

MODULE 3: Fourier Series Analysis of Signals (5 Lectures)

Response of LTI systems to complex exponential, representation of periodic signals: The Fourier series, properties of Fourier series, convergence of Fourier series.

MODULE 4: Fourier Transform Analysis of Signals (6 Lectures)

Representation of a-periodic signals: The Fourier Transform, properties of Fourier transform, System analysis by Fourier Transforms, convergence of Fourier transform.

MODULE 5: Sampling (5 Lectures)

Sampling theorem, effect of under-sampling, reconstruction of a signal from its samples, Spectrum of sampled signal.

MODULE 6: Z-Transform (6 Lectures)

Definitions, region of convergence, properties of Z-transform, inversion of Z-transforms, system function, applications to system analysis.

MODULE 7: Digital Filters (6 Lectures)

Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) systems, FIR and IIR filters, realization of FIR and IIR systems.

Suggested Text Books:

1. Oppenheim, A.V., Willsky, A.S., Nawab, S. H.: Signals and Systems, Prentice Hall India
2. .Rawat, T.K.: Signals and Systems, Oxford University Press3.Proakis, J.G.& Manolikis, D.G.: Digital Signal Processing-principles, algorithms and applications,Prentice Hall India

Reference Books:

1. Robert, M. J.: Signals and Systems, Tata McGraw Hill
2. Mitra, S.K.: Digital Signal Processing-a computer based approach, Tata McGraw Hill
3. Xavier, E: Signals, Systems & Signal Processing, S. Chand & Co.
4. Mastering MATLAB, Pearson Education (for Laboratory use).

Course Code	Course Title	Hours per week L-T-P	Credit
HS241406	**Finance and Accounting**	**3-0-0**	**3**

Prerequisites: High School Education
Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Develop a strong foundation in accounting principles, including the classification of accounts, the Double Entry System, the application of the Golden Rules of Debit and Credit, record financial transactions in journals, post them into ledgers, and prepare a Trial Balance.
- **CO2:** Understand the role of subsidiary books in accounting, including different types of cash
- books and their preparation; learn to reconcile differences between the Cash Book and Pass Book balances and prepare a Bank Reconciliation Statement
- **CO3:** Differentiate between capital and revenue expenditure, understand the concepts of bad debts and provisions for doubtful debts and gain the skills to prepare final accounts by incorporating necessary adjustments.
- **CO4:** Explore the concepts of savings and investment, including their benefits and differences, identify various investment avenues in physical and financial assets and understand the distinction between trading and investment to make informed financial decisions.
- **CO5:** Have a comprehensive understanding of financial markets, covering money and capital markets, key financial instruments, IPOs, stock exchanges, and mutual funds, and analyze the structure and functions of financial markets and assess different financial instruments used for investment.
- **CO6:** Acquire knowledge and skills in stock market investment and analysis, applying fundamental and technical analysis to evaluate stocks, learn to interpret stock market charts and patterns, assess stock price movements, and utilize various analytical tools for informed investment decisions.

MODULE 1: Introduction to Accounting
Concept and classification of Accounts, Transaction, Double Entry System of Book Keeping, Golden rules of Debit and Credit, Journal- Definition, Advantages, Procedure of Journalising, Ledger, Advantages, Rules regarding Posting, Balancing of ledger accounts, Trial Balance- Definition, Objectives, Procedure of preparation.

MODULE 2: Subsidiary Books, Cash Book and Bank Reconciliation
Name of Subsidiary Books, Cash Book- Definition, Advantages, Objectives, Types of Cash Book, Preparation of different types of Cash Books, Bank
Reconciliation Statement, Reasons of disagreement between Cash Book with Pass Book balance, Preparation of bank Reconciliation Statement.

MODULE 3: Expenditures, Bad Debts and Final Accounts
Concept of Capital Expenditure and Revenue Expenditure, Bad Debts,
Provision for Bad and Doubtful debts, Final Accounts: Preparation of Trading Account, Profit and Loss account and Balance Sheet with adjustments.

MODULE 4: Introduction to Saving and Investment
Savings: Concept, Benefits and saving alternatives, Disadvantage of Saving – Time Value of Money. Investment: Meaning, Advantage, Investment Avenues – Investment in Physical and financial assets, Difference between Saving and Investment; Trading- Meaning, Difference between Trading and Investment.

MODULE 5: Financial Market and Instruments.

Financial Markets and Instruments: Money Market, Capital Market, Money Market Instruments and features. Capital Markets and Features, New Issue Market and Stock Market; New Issue market- Initial Public Offering , Market Players, Secondary Market :Equity and Debentures, BSE and NSE, Index and its uses; Mutual Funds: Meaning and Types.

MODULE 6: Stock Market Investment and Analysis

Stock Market Investment: Fundamental Analysis- Economic Analysis-Industry Analysis and Technical Analysis - Tools of technical analysis, Understanding various types of charts and patterns- (Line chart, Bar Chart, Candlestick Chart Point and Figure Chart)

Textbooks/Reference Books

1. Dam, B. B.Sarda, R.A. Barman, R. Kalita, B., Theory and Practice of Accountancy- Vol. I, Gayatri Publication, 2019
2. Goel, D.K Goel, R. Goel,S. Accountancy, Avichal publishing company, 2020
3. Kevin, S., Security Analysis and Portfolio Management, Prentice Hall India., 2022
4. Pathak,B., Indian Financial System, Pearson Education, 2024
5. Chandra, P., The Invesment Game: How To win, Tata McGrawHill, 2012

Course Code	Course Title	Hours per week L-T-P	Credit
AU241407	**Environmental Science**	**3-0-0**	**0**

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Relate environment and its associated problems
- **CO2:** Explain basic ethics of living and protection of other organisms
- **CO3:** Applying sustainable development principles

MODULE 1: Introduction
Environment and Ecology, Objectives of ecological study, Aspects of Ecology, Autecology, Synecology; Ecosystem, Structural and functional attributes of an ecosystem, Food chain and foodweb, Energy flow, Bio geochemical cycles

MODULE 2: Land: Use and Abuse
Land use: Impact of land–use on environmental quality, Land degradation, Control of land degradation, Wasteland, Wet lands

MODULE 3: Water Pollution
Introduction: Water quality standards, Water pollution

MODULE 4: Air Pollution
Introduction, Air pollution system, Air pollutants, Air pollution laws

MODULE 5: Noise Pollution
Sources of noise pollution, Effects of noise, Physical effects, Physiological effects

MODULE 6: Control of Pollutions
Control of water pollution and management, Water pollution legislations, Water quality management in surface water; Control of air pollution, source correction method, Pollution control equipment; controls of Noise pollution

Textbooks/Reference Books

1. Dr Suresh Dhameja, Environmental engineering and management, 2010
2. Dr B.S. Chauhan, Environmental studies
3. Henry and Hence, Environmental science and engineering
4. Dr Susmitha Baskar, Environmental studies for undergraduate course
5. Clair Sawyer, Chemistry for environmental engineering and science

Course Code	Course Title	Hours per week L-T-P	Credit
PE241411	**Transducers and Sensors Lab**	**0-0-2**	**1**

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1**: Experiment with Passive transducers and draw the characteristics curve
- **CO2**: Experiment with transducers use for temperature, pressure, force, torque, speed measurements and draw the characteristics curve.
- **CO3**: Experiment with optical transducers and draw the characteristics curve
- **CO4**: Experiment with the response of first order system with test signals
- **CO5**: Experiment with active transducers and draw its characteristics curve

LIST OF EXPERIMENTS

1. To study the characteristics of rotary potentiometer and find its sensitivity.
2. To study the characteristics of thermistor and thermocouple.
3. To study the step response characteristic of RTD and thermocouple.
4. To study the temperature measurements using RTD with three and four leads.
5. To study the characteristics of optical transducers.
6. To study the output response analysis of first order system with standard test signals.
7. LVDT:
    a. To Study the characteristics of LVDT.
    b. To find its Linearity Range.
    c. To find its Sensitivity.
8. Strain gauge:
    a. To Study the characteristics of strain gauge
    b. To find its Sensitivity.
    c. To find its percentage Linearity.
9. SYNCHRO:
    a. To Study the characteristics of synchro transmitter and receiver.
    b. To Study about the torque synchro.
    c. To Study about its error detector.
10. To design LabVIEW VI for measurement of voltage, current and PQ.
11. To study the characteristics of piezoelectric sensors.
12. To study the operation of DAQ system for application with sensor signals.
13. To Design and develop sensing elements for sensing physical quantities, including biomedical signals, in electrical form.
14. Modeling & Simulation (Virtual Lab)
    a. Simulate the performance of a bio-sensor
    b. Measurement of level in a tank using capacitive type level probe
    c. Design an orifice plate for a typical application
    d. Simulate the performance of a chemical sensor

Course Code	Course Title	Hours per week L-T-P	Credit
PE241412	**Analog Circuits Lab**	**0-0-2**	**1**

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1**: Develop the ability to design, analyze, and simulate circuits using BJT and Op-Amp, including amplifiers, feedback circuits, oscillators, and filters.
- **CO2**: Evaluate the performance of different BJT amplifiers, including their gain, impedance, bandwidth, and efficiency.
- **CO3**: Implement and create various feedback amplifier circuits and design oscillators to calculate their frequency.
- **CO4**: Build and test practical circuits like summing amplifiers, integrators, differentiators, and waveform generators, applying what you learn through hands-on projects and simulations.

List of Laboratory Experiments/Demonstrations:

1. Introduction to Simulink/Multisim Study the biasing techniques of single stage BJT amplifiers (Fixed Bias and Voltage divider Bias) inSimulink/Multisim platform (take home assignment)
2. Find the β of the transistor.
3. Design and set up of the BJT common emitter amplifier using voltage divider bias with feedback and determine the gain-bandwidth product from its frequency response.
4. Realize BJT Darlington Emitter follower with and without bootstrapping and determine the gain, input and output impedances.
5. Set-up and study theworking of complementary symmetry class B push pull power amplifier and calculate the efficiency.
6. To design a BJT voltage series /current series/voltage shunt feedback amplifier circuit and calculate the following parameters with and without feedback.
    a. Mid band gain.
    b. Bandwidth and cut-off frequencies.
    c. Input and output impedance.
7. Design and set-up of the following tuned oscillator circuits using BJT, and determine the frequency of oscillation. Hartley Oscillator orColpitts Oscillator.
8. To study Op-Amp as inverting and non-inverting amplifier.
9. To study summing amplifier, integrator and differentiator circuits using op-amp.
10. To generate a triangular wave using IC741 op-amp.
11. To study active low pass and high pass filter using IC 741op-amp.
12. Lab project using Simulink/Multisim(for the project submission).

NB:

1. **It is advised to the students to design the circuits using SPICE before coming to the lab. This may be checked by the lab instructor.**
2. **Lab project using SPICE would be submitted on or before semester end. Choice of the lab projects may be finalized as per the mutual consent of the lab instructors and students. Lab projects may be started by students anytime in the semester as per the consent of the lab instructor.**

Course Code	Course Title	Hours per week L-T-P	Credit
PE241413	**Electrical Machines Lab**	**0-0-2**	**1**

Course Outcomes: After successful completion of the course, the students will be able to

- **CO1**: Develop the ability to perform tests on DC generators and motors, including O.C.C., load tests, and speed control of DC shunt motors.
- **CO2**: Analyze and perform test single-phase transformers through open circuit, short circuit, and load tests, and identify transformer connections.
- **CO3**: Perform and analyze no-load and blocked-rotor tests on 3-phase induction motors and Sumpner's test on single-phase transformers.
- **CO4**: Investigate the performance characteristics of DC machines, synchronous motors, and alternators through different tests.

Electrical Machines Lab-I

1. O.C.C. of D.C. generators
2. Load Test on D.C. shunt generators
3. Speed Control of D.C. shunt motors
4. Open circuit and short circuit test on single-phase transformers
5. Load test on single-phase transformers
6. Transformer Connections

Electrical Machines Lab-II

1. No-load and blocked-rotor tests on 3-phase induction motors Sumpner's Test or back-to-
2. back test on two similar single-phase transformers
3. Retardation test on a D.C. Machines
4. Hopkinson's or Back-to-back test on two similar D.C. Machines
5. Induction Regulator
6. V-Curves of a synchronous motor
7. Slip-Test on Alternators
8. Synchronization of alternators

Course Code	Course Title	Hours per week L-T-P	Credit
PR241415	**Micro Project (Skill Based)**	**0-0-4**	**2**

Prerequisites: Knowledge of basic science

Objectives:

- To identify problems based on societal or research needs
- To acquaint with the process of solving the problem in a group
- To apply engineering knowledge to attempt solutions to the problems

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Analyze engineering problems and their solutions
- **CO2:** Apply modern engineering tools to attempt solutions to the problems
- **CO3:** Adapt with computational thinking and automation skills
- **CO4:** Realize the need of professional communications, team work and self-learning

General Guidelines:

1. Students shall form a group of 3 to 4 students.
2. Each group shall conduct field survey or literature survey in consultation with faculty supervisor/mentor or internal project committee of faculty members to identify the needs of the society.
3. Each group shall formulate the problems on the basis of their survey and submit the implementation plan to the concerned faculty supervisor/mentor or the committee.
4. A log book is to be maintained by each group, wherein the group can record the work progress. The supervisor shall monitor the progress.
5. The faculty supervisor/mentor or the committee may give inputs to the students during the project activities, however, focus shall be on self-learning.
6. Students in a group shall understand problem effectively, propose multiple solutions and select the best solution in consultation with faculty supervisor/mentor or the committee.
7. Students shall convert the best solution into a working model/prototype using various components of their domain areas.
8. Students shall demonstrate and validate the model with proper justification and a report is to be compiled in a standard format/template

Assessment Criteria:

The Micro Project shall be assessed based on the following criteria:

Sl No.	Component	Weightage
1	Proper identification of the problem	20
2	Quality of survey	15
3	Clarity of problem definition	10
4	Presence of innovation	5
5	Effective use of modern engineering tools	10
6	Cost effectiveness	5
7	Societal impact	10
8	Functioning of the working model	15
9	Fulfilment of standard engineering norms	5
10	Clarity in written and oral communication	5
		100

Examinations:

Sl No.	Type of examination/evaluation	Marks (50)
1	Mid-term presentation on Progress	10
2	Demonstration of the project	10
3	End-term presentation	25
4	Viva-voce examination	5

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